FACULTY OF ENGINEERING TECHNOLOGY

Subject: comm.& Data Transmission — Mid Exam Summer 23/24

Name: — Duration: 50 min

This exam contains four questions

Q1:(Please write your answers into table only, otherwise your answers will be neglected) **(12 marks) (CLIO:1,7)**

Question	1	2	3	4	5	6	7	8
Answer								

1. The two criteria: throughput & delay are often contradictory, and this is due to:
   a) traffic b) traffic congestion c) traffic delay

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2. Baseband transmission is:
   a) a single channel system that support single transmission
   b) multi-channel system that support single transmission
   c) a single channel system that support multi -transmission

3. A measure of the capacity of a circuit or channel:
   a) SNR b) bandwidth c) bps

4. The value of the upper frequency of voice channel in terms of wavelength is:
   a) 75 km b) 7.5 km c) 75 m

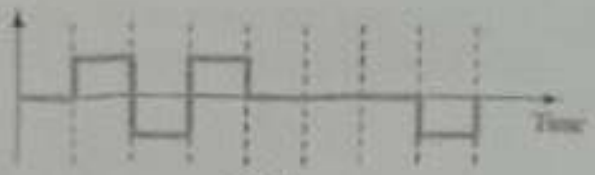
5. The following network is circuit switched network:
   a) Internet b) satellite c) telephony

6. Voice broadcasting mesh network contains 5 stations, the number of links in this network are:
   a) 10 b) 5 c)20

7. In a digital transmission, the sender clock is 0.2 percent faster than the receiver clock. How many extra bits per second does the sender send if the data rate is 1 Mbps?
   a) 2000 bits b) 200 bits c) 20 bits

8. Find the 8-bit data stream for each case depicted in Figure:

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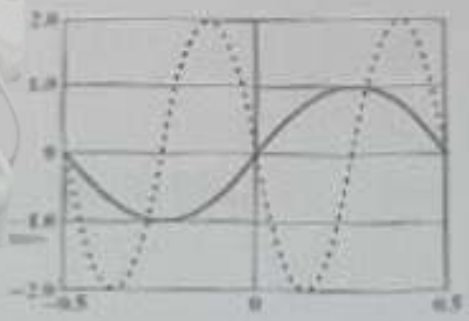
c. AMI

a) **01110001.** b) **11000100** c) **10011001.**

Q2: A TCP segment consisting of 1500 bits of data and 160 bits of header is sent to the IP layer, which appends another 160 bits of header. This is then transmitted through two networks, each of which uses a 24-bit packet header. The destination network has a maximum packet size of 800 bits. How many bits, including headers, are delivered to the network layer protocol at the destination? (5 marks) (CLIO:5)

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Q3: If the solid curve in Figure represents $\sin(2\pi f)$, what does the dotted curve represent? Write the equation of dotted curve in the form $A \sin(2\pi ft + \Phi)$; what are A, f, and Φ? (8 marks) (CLIO:9)



Q4: Assume that the TV picture is to be transmitted over a channel with 4.5-MHz bandwidth and a 35-dB signal-to-noise ratio. Find the capacity of the channel (bps).

(5 marks) (CLIO:9)



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